

Ejercicio Cifrado Afin

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6CV4

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4:50 PM

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Cifrado

$$k = (5, 25)$$

M = A R E L I

Z₂₇: 0 18 4 11 8

C = Y H R Z L

$$C_0 = (5 \cdot 0 + 25) \bmod 27 = 25 \text{ Y}$$

$$C_1 = (5 \cdot 18 + 25) \bmod 27 = 7 \text{ H}$$

$$C_2 = (5 \cdot 4 + 25) \bmod 27 = 18 \text{ R}$$

$$C_3 = (5 \cdot 11 + 25) \bmod 27 = 26 \text{ Z}$$

$$C_4 = (5 \cdot 8 + 25) \bmod 27 = 11 \text{ L}$$

Descifrado

$$k = (5, 25) \quad 5^{-1} = 11$$

C = Y H R Z L

Z₂₇: 25 7 18 26 11

M =

$$m_0 = (25 - 25)11 \bmod 27 = 0 \text{ A}$$

$$m_1 = (7 - 25)11 \bmod 27 = 18 \text{ R}$$

$$m_2 = (18 - 25)11 \bmod 27 = 4 \text{ E}$$

$$m_3 = (26 - 25)11 \bmod 27 = 11 \text{ L}$$

$$m_4 = (11 - 25)11 \bmod 27 = 8 \text{ I}$$

$$k = (5, 7) \quad 5^{-1} = 11$$

C = H L Z B L

Z₂₇: 7 11 26 1 11

M = A Q T O Q

$$m_0 = (7 - 7)11 \bmod 27 = 0 \text{ A}$$

$$m_1 = (11 - 7)11 \bmod 27 = 17 \text{ Q}$$

$$m_2 = (26 - 7)11 \bmod 27 = 20 \text{ T}$$

$$m_3 = (1 - 7)11 \bmod 27 = 15 \text{ O}$$

$$m_4 = (11 - 7)11 \bmod 27 = 17 \text{ Q}$$

